

Bounded Global Closure in Quartic Zero Repair

Local Exactness, Nonzero Closure Defect, and the Logarithmic Upper Bound

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Abstract

For the quartic zero-repair pair

$$P = \left(\frac{1}{8}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{8} \right), \quad Q = \left(\frac{7}{64}, \frac{5}{16}, \frac{5}{32}, \frac{5}{16}, \frac{7}{64} \right),$$

satisfying

$$G_P(z) - G_Q(z) = \frac{(1-z)^4}{64},$$

the previously proposed zero-loss common-anchor mechanism appears to be too strong at finite horizons.

Exact finite-support linear programs show instead a different structure: all local zero-repair identities close to numerical precision, while a strictly positive global common-anchor defect remains.

This defect is remarkably stable under support enlargement and appears to remain bounded as the horizon increases.

We isolate the resulting conjecture:

$$\sup_{n \geq 1} \left[\mathcal{A}_n \left(\frac{3}{20} \right) - \mathcal{B}_n \right] < \infty.$$

Here $\mathcal{B}_n$ is the optimal depth- n bill and $\mathcal{A}_n(\varepsilon)$ is the least root bill among exact depth- n trees admitting one common depth- n viable anchor after Bernoulli insurance

$$R_\varepsilon = (1 - \varepsilon)\delta_0 + \varepsilon\delta_1.$$

We prove that any uniform bound on this global closure defect implies

$$\mathcal{B}_{2n} \leq \mathcal{B}_n + C + \frac{3}{20},$$

and therefore

$$\mathcal{B}_n = O(\log n).$$

Thus the logarithmic upper bound does not require the global defect to vanish. It requires only that the defect remain uniformly bounded.

1 Quartic zero repair

Let

$$\mathcal{A} = \{0, 1, 2, 3, 4\},$$

and let P, Q be the laws

$$P = \left(\frac{1}{8}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{8} \right), \quad Q = \left(\frac{7}{64}, \frac{5}{16}, \frac{5}{32}, \frac{5}{16}, \frac{7}{64} \right).$$

Their generating functions satisfy the exact quartic identity

$$G_P(z) - G_Q(z) = \frac{(1-z)^4}{64}.$$

A depth- n zero-repair tree is a family

$$\{B_h\}_{|h| \leq n}$$

of probability laws on $\mathbb{Z}_{\geq 0}$, indexed by words $h \in \mathcal{A}^{|h|}$, satisfying

$$Q * B_h = \sum_{x \in \mathcal{A}} P(x) \delta_x * B_{hx}, \quad |h| < n. \quad (\text{ZR})$$

Let K_n denote the set of all laws occurring as roots of exact depth- n zero-repair trees. The optimal depth- n bill is

$$\mathcal{B}_n := \inf_{B \in K_n} \mathbb{E}B.$$

The sequence $(\mathcal{B}_n)$ is nondecreasing.

2 The sharp Bernoulli insurance law

For $0 \leq \varepsilon \leq 1$, define

$$R_\varepsilon = (1 - \varepsilon)\delta_0 + \varepsilon\delta_1.$$

For the quartic pair, the sharp threshold in this Bernoulli family is

$$Q * R_\varepsilon \succeq_{\text{st}} P \iff \varepsilon \geq \frac{3}{20}.$$

Hence the distinguished reserve is

$$R_* = \frac{17}{20}\delta_0 + \frac{3}{20}\delta_1, \quad \mathbb{E}R_* = \frac{3}{20}.$$

Moreover, convolution preserves stochastic order, so for every probability law B ,

$$Q * (B * R_*) \succeq_{\text{st}} P * B. \quad (1)$$

The viability sets are stochastically upward closed:

$$C \in K_n, \quad D \succeq_{\text{st}} C \implies D \in K_n. \quad (2)$$

These two facts are the algebraic basis of the common-anchor construction.

3 Common-anchor optimization

The original common-anchor mechanism asks whether a near-optimal depth- n tree can be chosen so that all last internal nodes admit one common depth- n viable anchor after insurance by R_* .

To separate this global requirement from ordinary depth- n optimality, we introduce an explicit value function.

Definition 1 (Anchored bill). *For $0 \leq \varepsilon \leq 1$, let*

$$\mathcal{A}_n(\varepsilon)$$

be the infimum of $\mathbb{E}B_\varnothing$ over all exact depth- n zero-repair trees

$$\{B_h\}_{|h| \leq n}$$

for which there exists a single law

$$C_n \in K_n$$

such that

$$\boxed{(Q * B_h) * R_\varepsilon \succeq_{\text{st}} P * C_n \quad \forall |h| = n - 1.} \quad (3)$$

Clearly

$$\mathcal{A}_n(\varepsilon) \geq \mathcal{B}_n.$$

This motivates the central quantity.

Definition 2 (Global closure defect). *Define*

$$\boxed{\mathcal{G}_n(\varepsilon) := \mathcal{A}_n(\varepsilon) - \mathcal{B}_n.} \quad (4)$$

The zero-loss common-anchor conjecture is exactly the assertion

$$\mathcal{G}_n(3/20) = 0$$

for every n , up to arbitrarily small near-optimality loss.

The numerical evidence below indicates that this condition is unnecessarily strong.

4 Local exactness versus global closure

The distinction revealed by the finite-horizon computations is:

$$\boxed{\text{local zero-repair identities close exactly, while global common closure costs extra bill.}}$$

At every internal node, the equality

$$Q * B_h = \sum_x P(x) \delta_x * B_{hx}$$

is imposed exactly.

In the numerical solutions, the corresponding equality residuals are at floating-point precision.

Nevertheless,

$$\mathcal{G}_n(3/20) > 0$$

already for small horizons.

Thus the global defect is not a local violation of the zero-repair law.

It is a nonlocal compatibility cost created by requiring *one common viable anchor simultaneously below all insured last-layer parents*.

This distinction is the relevant mathematical analogue of a general local-versus-global closure phenomenon:

$$\boxed{\text{vanishing local residual} \not\Rightarrow \text{vanishing global closure term.}}$$

No physical interpretation is required for the argument below.

5 Finite-support evidence

The DREAM6-IMBA closure solver imposes the exact depth- n recursion on laws supported on

$$\{0, 1, \dots, M\}.$$

For each n, M , it computes the finite-support values

$$\mathcal{B}_n^{(M)}$$

and

$$\mathcal{A}_n^{(M)}(3/20),$$

and therefore

$$\mathcal{G}_n^{(M)}(3/20) = \mathcal{A}_n^{(M)}(3/20) - \mathcal{B}_n^{(M)}.$$

The first stable values are:

n	M	$\mathcal{B}_n^{(M)}$	$\mathcal{G}_n^{(M)}(3/20)$
2	20	0.238888888888889	0.038518518518519
2	30	0.238888888888889	0.038518518518519
2	40	0.238888888888889	0.038518518518519
3	20	0.296480916228441	0.063673488949131
3	30	0.296480916228441	0.063673488949128
3	40	0.296480916228441	0.063673488949134
4	20	0.344825164056711	0.065942286885605

For $n = 2$, the closure defect is exactly recognized numerically as

$$\boxed{\mathcal{G}_2^{(M)}(3/20) = \frac{26}{675}}$$

for all tested

$$M = 20, 30, 40.$$

The support boundary mass is numerically zero.

For $n = 3$, the closure defect is again stable to the displayed precision across

$$M = 20, 30, 40,$$

while the boundary mass is at numerical noise level.

The $n = 4, M = 20$ run has larger numerical residuals and small cutoff-scale mass, and should therefore be treated as provisional until larger support values confirm stability.

The most conservative conclusion from the data is therefore:

$$\boxed{\mathcal{G}_2(3/20) > 0, \quad \mathcal{G}_3(3/20) > 0}$$

at least at the level of strongly support-stable finite computations.

Thus the zero-loss common-anchor mechanism is not supported by these experiments.

6 Tail-layer interpretation

Let J_0 denote the root reserve under an anchored optimizer and define the tail coordinates

$$T_k = \Pr(J_0 \geq k).$$

For every nonnegative integer-valued law,

$$\mathbb{E}J_0 = \sum_{k \geq 1} T_k.$$

Hence the global closure defect admits the exact decomposition

$$\boxed{\mathcal{G}_n = \sum_{k \geq 1} \left(T_k^{\text{anch}}(n) - T_k^{\text{base}}(n) \right)}. \quad (5)$$

At $n = 2$, both the baseline and anchored root laws are supported on $\{0, 1\}$, so the entire closure defect lies in the first tail layer:

$$\mathcal{G}_2 = \Delta T_1.$$

At $n = 3$, the anchored optimizer develops a nonzero second layer

$$\Pr(J_0 = 2) \approx 0.00168410518.$$

Thus

$$\mathcal{G}_3 = \Delta T_1 + \Delta T_2.$$

At $n = 4$, the higher-layer contribution is substantially larger.

This suggests that the global closure bill is not generated by a single scalar correction.

Rather, the closure mass is redistributed among progressively activated tail layers.

The numerical picture is therefore consistent with a decomposition of the form

$$\boxed{T^{\text{new}} = M_n T^{\text{old}} + r_n}, \quad (6)$$

where the internal transport M_n approximately preserves total tail mass,

$$\mathbf{1}^\top M_n = \mathbf{1}^\top,$$

while

$$\mathbf{1}^\top r_n$$

is the genuinely new global closure contribution.

Equation (6) is presently a structural hypothesis, not a proved identity.

7 The bounded global closure principle

The key observation is that the logarithmic upper bound does not require

$$\mathcal{G}_n(3/20) = 0.$$

It requires only a uniform bound.

Conjecture 3 (Bounded Global Closure). *There exists a constant $C_0 < \infty$ such that*

$$\boxed{\mathcal{G}_n\left(\frac{3}{20}\right) = \mathcal{A}_n\left(\frac{3}{20}\right) - \mathcal{B}_n \leq C_0 \quad \forall n \geq 1.} \quad (\text{BGC})$$

This conjecture is strictly weaker than the zero-loss common-anchor conjecture.

It is nevertheless sufficient for the logarithmic upper bound.

8 Main theorem

Theorem 4 (Bounded global closure implies logarithmic growth). *Assume Conjecture (BGC). Then*

$$\boxed{\mathcal{B}_{2n} \leq \mathcal{B}_n + C_0 + \frac{3}{20} \quad \forall n \geq 1.} \quad (7)$$

Consequently,

$$\boxed{\mathcal{B}_n = O(\log n).}$$

Proof. Fix $n \geq 1$ and $\eta > 0$.

By the definition of $\mathcal{A}_n(3/20)$, choose an exact depth- n zero-repair tree

$$\{B_h\}_{|h| \leq n}$$

and a common anchor

$$C_n \in K_n$$

such that

$$\mathbb{E}B_\emptyset \leq \mathcal{A}_n\left(\frac{3}{20}\right) + \eta$$

and

$$(Q * B_h) * R_* \succeq_{\text{st}} P * C_n \quad \forall |h| = n - 1.$$

Convolve every prefix law above the last internal layer by the same reserve:

$$\widehat{B}_h = B_h * R_*, \quad |h| \leq n - 1.$$

For

$$|h| \leq n - 2,$$

the original exact zero-repair identity gives

$$Q * B_h = \sum_x P(x)\delta_x * B_{hx}.$$

Convolution by the same R_* therefore preserves the identity:

$$Q * \widehat{B}_h = \sum_x P(x)\delta_x * \widehat{B}_{hx}.$$

Thus all previously constructed local zero-repair equalities remain exact.

Now fix a last internal history

$$|h| = n - 1.$$

By the common-anchor condition,

$$(Q * B_h) * R_* \succeq_{\text{st}} P * C_n.$$

By Strassen's theorem, one may couple

$$S \sim (Q * B_h) * R_*$$

and

$$X + J \sim P * C_n$$

so that

$$S \geq X + J \quad \text{a.s.}$$

Conditioning on $X = x$, define

$$D_{hx} = \text{Law}(S - X \mid X = x).$$

Then

$$D_{hx} \succeq_{\text{st}} C_n.$$

Since

$$C_n \in K_n$$

and K_n is stochastically upward closed,

$$D_{hx} \in K_n.$$

Therefore every depth- n child can be extended by another exact depth- n zero-repair tree. This produces an exact depth- $2n$ tree whose root law is

$$B_{\varnothing} * R_*.$$

Hence

$$\begin{aligned} \mathcal{B}_{2n} &\leq \mathbb{E}B_{\varnothing} + \mathbb{E}R_* \\ &\leq \mathcal{A}_n\left(\frac{3}{20}\right) + \eta + \frac{3}{20}. \end{aligned}$$

By Conjecture (BGC),

$$\mathcal{A}_n\left(\frac{3}{20}\right) \leq \mathcal{B}_n + C_0.$$

Thus

$$\mathcal{B}_{2n} \leq \mathcal{B}_n + C_0 + \frac{3}{20} + \eta.$$

Letting $\eta \downarrow 0$ gives

$$\boxed{\mathcal{B}_{2n} \leq \mathcal{B}_n + C_0 + \frac{3}{20}.}$$

Iterating on dyadic horizons,

$$\mathcal{B}_{2^k} \leq \mathcal{B}_1 + k \left(C_0 + \frac{3}{20} \right).$$

For arbitrary n , choose

$$k = \lceil \log_2 n \rceil.$$

By monotonicity,

$$\mathcal{B}_n \leq \mathcal{B}_{2^k} \leq \mathcal{B}_1 + \left(C_0 + \frac{3}{20} \right) \lceil \log_2 n \rceil.$$

Therefore

$$\boxed{\mathcal{B}_n = O(\log n).}$$

□

9 A numerical candidate

The observed closure defects are

$$0.0385185, \quad 0.0636735, \quad 0.0659423.$$

A simple rational upper scale suggested by these values is

$$\frac{1}{15} = 0.06666666\ldots$$

This motivates a stronger quantitative conjecture.

Conjecture 5 (One-fifteenth closure bound). *For the quartic zero-repair pair,*

$$\boxed{\mathcal{G}_n\left(\frac{3}{20}\right) \leq \frac{1}{15} \quad \forall n \geq 1.} \quad (C_{1/15})$$

At present this constant is suggested only by the first few finite-horizon computations. It is not derived from an exact identity and should not yet be regarded as canonical. Nevertheless, if Conjecture $(C_{1/15})$ holds, then

$$\mathcal{B}_{2n} \leq \mathcal{B}_n + \frac{1}{15} + \frac{3}{20}.$$

Since

$$\frac{1}{15} + \frac{3}{20} = \frac{4}{60} + \frac{9}{60} = \boxed{\frac{13}{60}},$$

we obtain

$$\boxed{\mathcal{B}_{2n} \leq \mathcal{B}_n + \frac{13}{60}.$$

Therefore

$$\boxed{\mathcal{B}_n \leq \mathcal{B}_1 + \frac{13}{60} \lceil \log_2 n \rceil.}$$

10 Critical-reserve viewpoint

The same global obstruction may be measured in a dual way.

Define the critical reserve

$$\varepsilon_{\text{crit}}(n)$$

as the smallest Bernoulli parameter for which the common-anchor condition can be achieved with zero additional root-bill loss.

Finite-support bisection gives approximately

$$\varepsilon_{\text{crit}}(2) \approx 0.215000,$$

$$\varepsilon_{\text{crit}}(3) \approx 0.264897,$$

and

$$\varepsilon_{\text{crit}}(4) \approx 0.304160$$

in the currently available runs.

For $n = 2$ and $n = 3$, these values are stable under the tested support increase.

Thus the same global closure obstruction can be paid in two equivalent numerical currencies:

keep $\varepsilon = \frac{3}{20}$
and pay $\mathcal{G}_n > 0$,

or

increase ε
and force $\mathcal{G}_n = 0$.

For the logarithmic upper bound, the first formulation is preferable: one needs only a uniform bound on $\mathcal{G}_n$, not zero defect.

11 What must now be proved

The numerical evidence suggests that the main open problem has shifted.

The natural target is no longer

$$\mathcal{G}_n(3/20) = 0.$$

It is

$$\sup_n \mathcal{G}_n(3/20) < \infty.$$

There are several possible routes.

1. Direct primal bound

Construct, for every n , an anchored depth- n tree satisfying

$$\mathbb{E}B_\emptyset \leq \mathcal{B}_n + C_0$$

with a constant C_0 independent of n .

2. Tail-layer transport

Show that the additional anchored reserve admits a decomposition

$$T^{\text{anch}}(n) = M_n T^{\text{base}}(n) + r_n$$

with

$$\mathbf{1}^\top M_n = \mathbf{1}^\top$$

and

$$\sup_n \mathbf{1}^\top r_n < \infty.$$

Then

$$\mathcal{G}_n = \mathbf{1}^\top r_n$$

is uniformly bounded.

3. Dual certificate

Extract a finite family of common-anchor dual inequalities whose optimal value is bounded uniformly in n .

The exact repeated dual patterns observed by DREAM6-IMBA are natural candidates for such a certificate.

4. Support-independent compactness

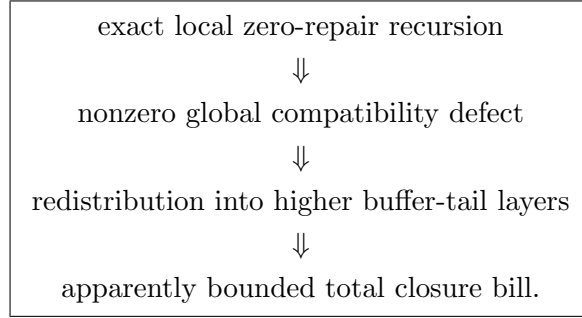
Convert the stable finite-support calculations into unrestricted statements by proving that sufficiently near-optimal anchored solutions cannot improve their objective through mass escaping to arbitrarily large buffer states.

This would justify passing

$$M \rightarrow \infty.$$

12 Interpretation of the current data

The current evidence supports the following structural picture:



The defect is therefore not a failure of the local recursion.

It is the price of simultaneous global closure.

This resolves an important conceptual ambiguity: one should not attempt to eliminate the global term by repeatedly refining the local representation.

The correct objective is to prove that the surviving global term is bounded.

Conclusion

The numerical common-anchor experiments modify the original conjectural picture in a precise way.

The strongest zero-loss claim,

$$\mathcal{A}_n\left(\frac{3}{20}\right) = \mathcal{B}_n,$$

does not agree with the first stable finite-support computations.

However, the quantity

$$\mathcal{G}_n\left(\frac{3}{20}\right) = \mathcal{A}_n\left(\frac{3}{20}\right) - \mathcal{B}_n$$

appears to be a small, support-stable, nonzero global closure defect.

The logarithmic upper bound requires only that this defect be uniformly bounded.

Thus the central open question is now

For quartic zero repair with the sharp Bernoulli reserve

$$R_* = \frac{17}{20}\delta_0 + \frac{3}{20}\delta_1,$$

is the additional root bill required to impose one common depth- n viable anchor bounded independently of n ?

Equivalently,

$$\sup_{n \geq 1} \left[\mathcal{A}_n \left(\frac{3}{20} \right) - \mathcal{B}_n \right] < \infty ?$$

A positive answer immediately gives

$$\mathcal{B}_{2n} \leq \mathcal{B}_n + C$$

for a universal constant C , and hence

$$\mathcal{B}_n = O(\log n).$$

The remaining problem is therefore not to make the global closure defect vanish. It is to prove that it cannot grow.